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RESEARCH PROPOSAL

BEYOND BEAUTY: WHAT REJECTED STONES STILL KNOW

Repurposing Gemstone Authentication Data to Identify Functional Value in Non-Gem-Grade Tourmaline

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RESEARCH SPECIFICATIONS

Field: Materials Science

Research Topic: Functional Applications of Non-Gem-Grade Tourmaline

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IARCO

Beyond Beauty: What Rejected Stones Still Know

Repurposing Gemstone Authentication Data to Identify Functional Value in Non-Gem-Grade Tourmaline

Introduction and Research Problem

My father has been cutting and grading gemstones for over twenty years. I've watched him hold a tourmaline up to the light more times than I can count, and within seconds, decide its fate: a case in a jewelry store, or a tray of rejects that never leaves the workshop. The criteria are always the same: clarity, color, and how light moves through the crystal. What I never considered, until researching tourmaline's structure for another project, is that the stone in the reject tray doesn't know it failed. Its underlying tourmaline structure remains capable of the same kinds of electromechanical and thermal responses as the one bound for a display case—still non-centrosymmetric, still able to generate a voltage when compressed or warmed, although composition and crystal quality may affect how strongly. The key distinction between the two stones comes down to human decisions about how they look.

That distinction between what a stone is and what it is judged to be is the starting point of this proposal.

Literature Review

Gemologists already run detailed chemical analysis on tourmaline, but for a different reason entirely. Trace-element fingerprinting is used to help determine where a stone came from and assess its authenticity, with laboratories such as GIA relying on quantitative chemical analysis for geographic-origin determination [1]. Materials scientists, separately, have established that elemental composition—particularly iron content—influences tourmaline's pyroelectric and piezoelectric performance [2]–[5]. These two fields have largely developed separately. To my knowledge, no systematic framework combines authentication datasets with functional physics to evaluate whether non-gem material holds unrecognized electromechanical value.

Connecting these areas requires linking crystal chemistry to physical properties. Bosi and Lucchesi [2] mapped how site occupancy governs structural variation across the tourmaline group, while van Hinsberg et al. [6] showed that composition records formation environments so faithfully that the mineral functions as a geological indicator. Exploring electrical output directly, Pandey and Schreuer [3] measured elastic and piezoelectric constants across temperature ranges, Hawkins et al. [5] evaluated compositional influences on pyroelectricity, and Whatmore [4] set out the physical basis of pyroelectric response in non-centrosymmetric crystals. Yet while these studies confirm that coefficients vary with composition, they rely on small sets of laboratory specimens rather than leveraging the trace-element data gemological labs generate at scale.

Application research provides necessary realistic boundaries. Reviews by Anton and Sodano [7] and Sezer and Koç [8] show that engineered piezoelectric ceramics dominate energy harvesting, bounding any output claims made here. Nevertheless, Han et al. [9] report that ultrafine non-gem tourmaline particles can improve nitrogen removal in wastewater reactors, suggesting that non-gem material has real-world functional value. That work, however, treats tourmaline as a bulk material, with no attempt to separate higher-performing stones from lower-performing ones.

This project sits right where these lines of research miss each other. Compositional data exists in quantity, the composition–performance relationship is established in principle, and a use case for non-gem material exists. What is missing is a framework that uses the first to predict the second in service of the third. This proposal addresses that gap.

This reflects what E.O. Wilson termed consilience—the idea that major progress often happens when separate disciplines realize they are describing the same reality in different terms [10]. Gemology and materials science have been extracting different meanings from the same crystal. In other words, a stone rejected by gemology may still be selected by materials science.

Research Question
Can trace-element compositional data already collected for gemstone authentication serve as a useful predictor of piezoelectric and pyroelectric performance in non-gem-grade tourmaline?

Hypothesis
Trace-element compositional data collected for gemstone authentication contain sufficient predictive information to enable preliminary screening of non-gem-grade tourmaline for piezoelectric and pyroelectric potential.

Research Design and Methodology
First, compile published composition–coefficient pairs from existing tourmaline literature, matching known compositional variables to measured piezoelectric and pyroelectric values across species (schorl, dravite, elbaite) [3], [5]. Second, analyze the relationship between elemental composition and reported coefficients using correlation analysis and multivariable regression where data permits. Third, compile existing compositional data from characterized reject-grade stones sourced directly from gem-cutting waste at my father's workshop in Tripoli, Lebanon; if certified records are unavailable for specific samples, those stones will be reserved for future physical validation. Fourth, apply the model to the sourced stones to estimate expected electrical output while evaluating how strongly compositional variables explain reported variation.

Rather than measuring physical device performance directly, this phase checks whether authentication data contains a strong enough predictive signal to justify future lab testing. A key limitation is that compositional data alone cannot capture all factors affecting electrical performance, making experimental validation an essential future step. If laboratory access is secured in later stages, a subset of predicted high- and low-performance samples will undergo physical testing to evaluate model accuracy.

Data Analysis

The analysis begins by treating the published dataset as a training set. Each compositional variable (Fe, Mn, Al, Mg content, and Fe/Mg ratio) will be tested individually against reported piezoelectric and pyroelectric coefficients using Pearson correlation to isolate predictive signals. Variables clearing a significance threshold will feed into a multiple linear regression, with the coefficient of determination (R^2) showing how much variance composition actually explains. An R^2 below roughly 0.4 would simply mean authentication data lacks sufficient predictive power, giving a clear negative answer to the research question.

Once fitted, I will run the model against the local reject stones to generate predicted coefficients and rank the candidate samples. Because physical measurements do not exist for these local stones yet, this step evaluates internal consistency—checking whether stones of identical species and similar composition yield logical, stable rankings. Cross-validation on the published dataset provides the accuracy estimate that the reject stones cannot yet supply. All data, code, and intermediate outputs will be preserved for full reproducibility.

Expected Outcome

The study will determine whether compositional variables consistently explain variation in tourmaline's piezoelectric and pyroelectric coefficients, demonstrating whether existing authentication data can support preliminary functional screening. A weak or inconsistent relationship would be equally informative, showing that trace-element data alone is insufficient for screening.

Significance of the Research Topic

Gem-grade tourmaline is valued primarily for beauty. Clarity and color determine market value, while potential functional properties are ignored. This proposal introduces a second axis of value running alongside the first: functional grade. A stone that fails aesthetically may still pass functionally; this proposal asks whether existing compositional data can identify such stones before they are discarded.

The significance extends beyond tourmaline. Gem-cutting workshops generate substantial quantities of rejected material with little or no commercial use. While non-gem tourmaline has a secondary life in environmental applications like wastewater treatment [9], no framework exists to screen which rejected stones are worth directing toward such uses. Screened reject tourmaline will not power infrastructure; its output per gram is far below engineered ceramics [7], [8]. Its realistic value lies in low-power applications: passive environmental sensors, small-scale monitoring devices, or niche energy-harvesting components where near-zero material cost matters more than raw output. More broadly, the project explores whether industrial byproducts classified as waste can be re-evaluated through functional rather than aesthetic criteria.

None of this requires new fieldwork or equipment at the proposal stage; the aim is simply to use existing data to ask a different question. My father discards stones by the tray. Somewhere in that tray may be material setting idle simply because no one thought to measure what it could do. The

information needed to recognize that value may already be sitting in existing certification records.

Practicalities

I built this project around what I can actually access. The correlation model relies on published literature, requiring only database access and spreadsheet software—no funding, no instrumentation, no institutional lab. The reject stones come from my father's workshop in Tripoli at no cost, and any existing certification records for them are already in his possession. Analysis will be performed in Excel and Python, both of which I use already. The only resource I do not currently have is piezoelectric testing equipment, which is why device-level measurement sits outside this proposal's current scope.

The project is scheduled across six months. Months 1–2: compile composition–coefficient pairs from published literature and build the reference dataset. Month 3: construct and internally check the correlation model. Month 4: source and catalog the 20–30 reject stones and match each to any available certification data. Month 5: apply the model to the sourced stones and assess how much reported variation composition explains. Month 6: write up results and prepare the manuscript for submission. If lab access is secured, experimental validation will extend the timeline by two months.

No human or animal participants are involved, so no ethical approval is required. The stones are industrial waste used with permission. All underlying data will be cited in full so the analysis can be independently verified.

Roadblocks and Potential Limitations

The main constraint is that published data comes from global deposits, whereas local samples come from a single region. Geologic variation means the correlation model may transfer imperfectly to Tripoli material. On a practical level, missing trace elements in older lab records may force me to exclude some stones, and the 20–30 sample size limits detailed subgroup analysis. Until physical testing equipment is secured, these predictions remain theoretical estimates rather than validated values.

Post-Program Plan

My first step after IARCO is to use reviewer feedback to tighten the model before attempting anything bigger. I plan to submit this proposal to Youth Research Journal. In parallel, the concrete next step is practical: going back to my father's workshop with a defined list of stones to screen. If compositional predictions hold up against what is already known about those stones, my next move is finding a lab that can run physical piezoelectric and pyroelectric tests on top candidates. What happens after that depends on what that first round of screening shows.

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